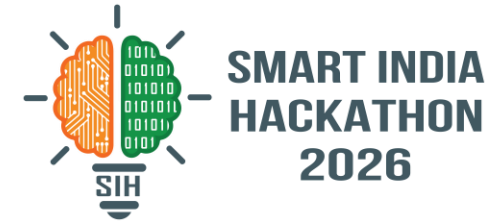
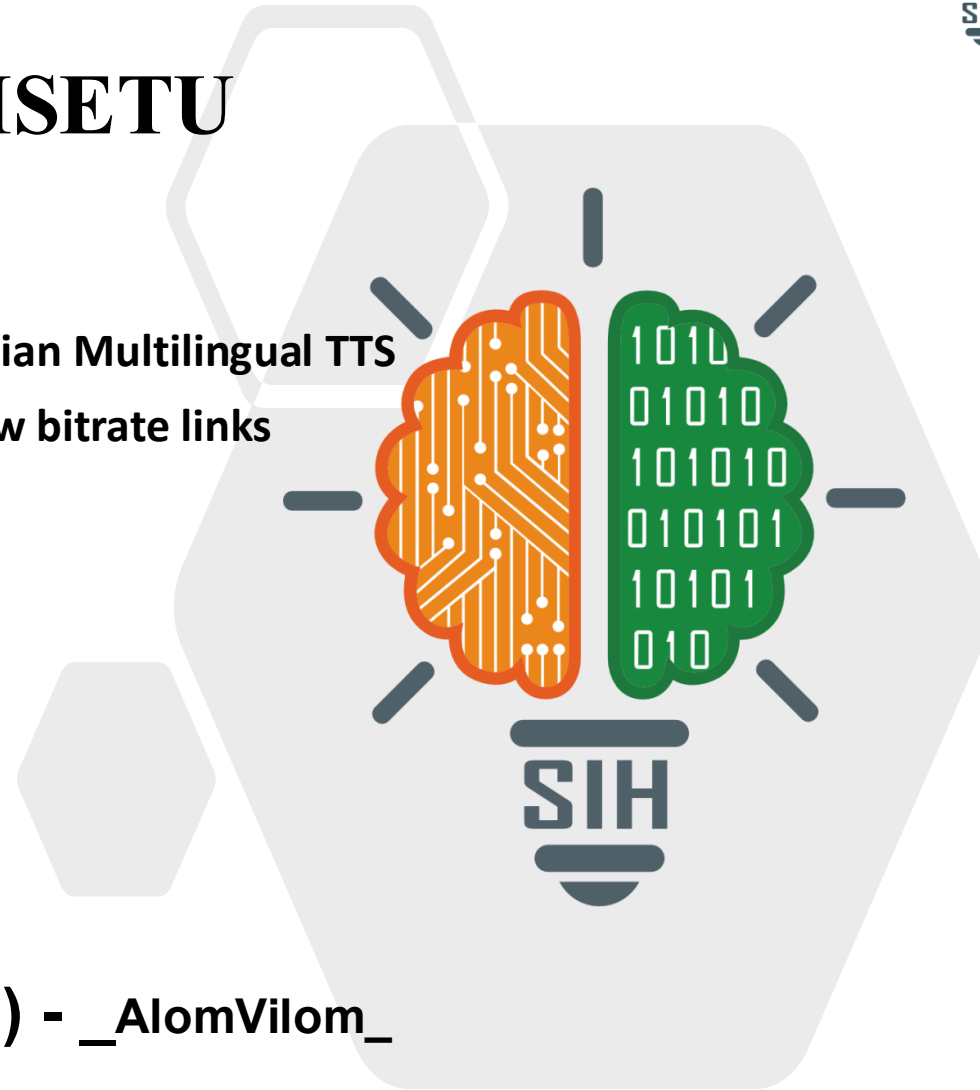


SMART INDIA HACKATHON 2026



VAANISETU

- **Problem Statement ID** – 26173
- **Problem Statement Title**- iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- NSUT062
- **Team Name (Registered on portal)** - __AlomVilom__



PROPOSED SOLUTION



Offline-first voice communication
for disaster and low-connectivity zones.



Speech is converted to **compact text on-device** before transmission.



10 Indian languages with on-device
IndicConformer ASR + IndicF5 TTS.



Push-to-talk relay mode for radio-like
communication.



SOS / distress alerts bypass silent
mode and DND.



Bluetooth SPP / Wi-Fi Direct for phone-
to-phone communication without
cellular towers.



Open-source, no cloud APIs, no
proprietary SDK dependency.



On-device ASR
IndicConformerASR
(multi-language
STT)



**10 Indian
languages**
Hindi + 9 others



**Compact
text relay**
Small text
packets over
local network



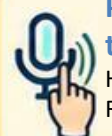
**IndicF5
offline TTS**
Natural voice
in Indian
languages



**Bluetooth
SPP**
Direct device to
device
communication



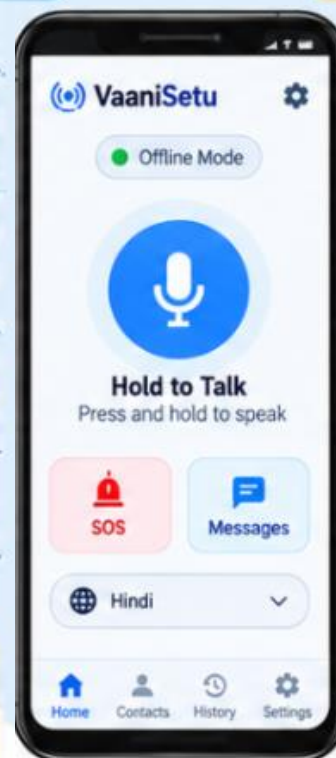
Wi-Fi Direct
Local high-
speed
connection (no
internet)



**Push-to-
talk**
Hold to speak
Release to
send



**SOS /
Distress
alert**
Bypass silent
mode and DND



"No tower. No cloud. Still connected."

IT ADDRESSES THE PROBLEM BY

- Works when **cellular towers** or **internet** are **unavailable**.
- **Reduces bandwidth** by transmitting text instead of raw voice.
- Enables **multilingual** safety-critical communication.
- Supports **rapid PTT relay** and emergency alerts.
- **Fully offline** and **lightweight** for field deployment.

TECHNOLOGY / PLATFORM



Android
(Kotlin/Java)



Bluetooth
(SPP)



Wi-Fi Direct

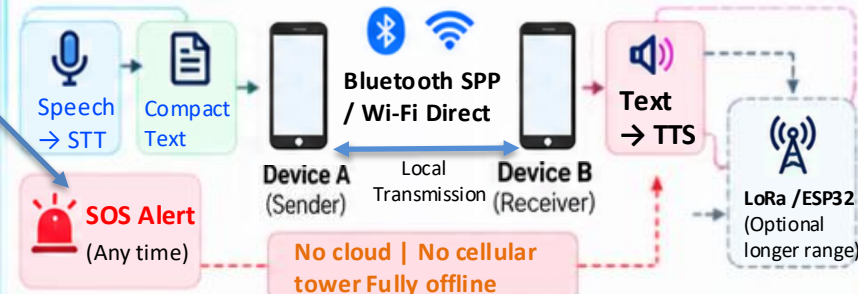


On-device AI
(IndicConformerASR
+ IndicF5 + VAD)



LoRa / ESP32
(Optional)

SYSTEM WORKFLOW (DEVICE TO DEVICE)



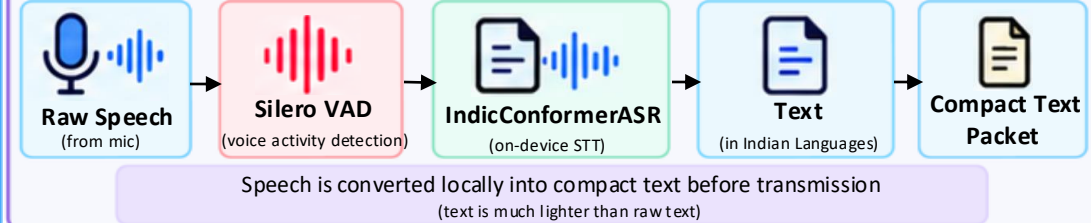
TECH STACK



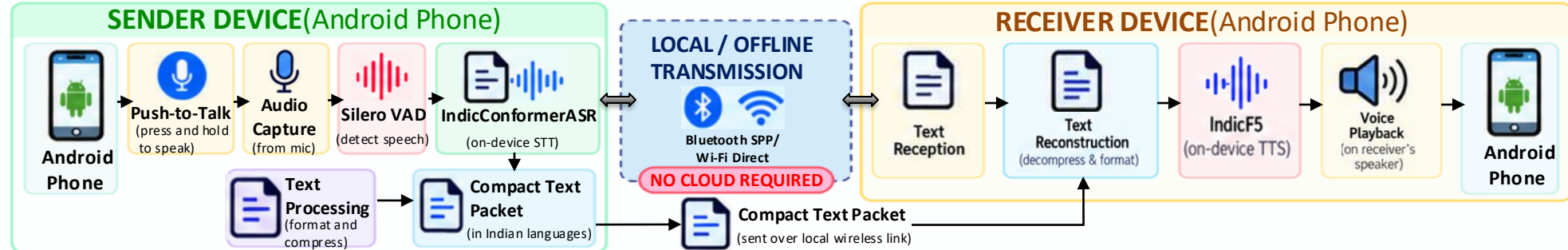
WORKING PROTOTYPE : [Click Here](#)



DATA TRANSFORMATION



SYSTEM FLOW (OFFLINE COMMUNICATION)



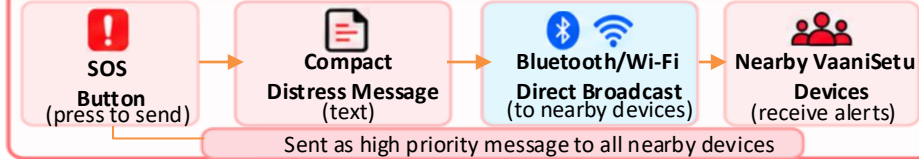
KEY OPEN SOURCE REPOSITORIES

-  **AI4Bharat IndicConformer**
Indic ASR and TTS for Indian languages [Click Here](#)
-  **VITS / IndicF5**
Neural TTS for Indian languages [Click Here](#)
-  **Sherpa-ONNX (k2-fsa)**
Efficient ASR inference [Click Here](#)
-  **ONNX Runtime Mobile**
Optimized for mobile and low-resource environments [Click Here](#)
-  **Silero VAD (MIT)**
Lightweight voice activity detection [Click Here](#)

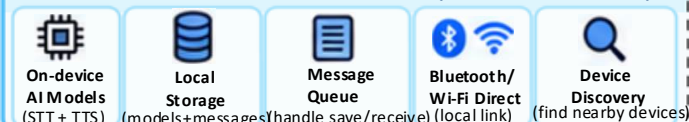
USER FEATURES



EMERGENCY / SOS (High Priority Path)



LOCAL INFRASTRUCTURE (On Each Device)



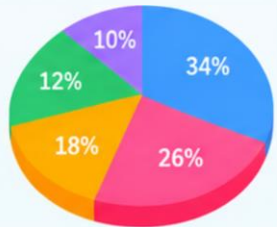
- FULLY OFFLINE
- NO CLOUD APIs
- LOCAL INFERENCE

PLANNED/EXTENSION (Longer Range Communication)



LoRa / ESP32
(for longer range communication in remote areas)

PLANNED/OPTIONAL

FINANCIALS(Estimated)**Android App Development****₹40K-₹70K**UI/UX | Audio API
Testing | Optimization**On-device AI models****₹30K-₹50K**Model integration
Optimization
10 Indian Languages**Communication module****₹20K-₹35K**Bluetooth SPP Wi-Fi
Direct Testing**Hardware-Extension(Optional)****₹25K-₹40K**LoRa/ESP32
(Prototype)
Long-Range testing**Others****₹15K-₹25K**Documentation | User Testing
Deployment | Contingency**Total Estimated cost****₹1.3-2.0 Lakhs****COST DISTRIBUTION**

■ Android App Development	34%
■ On-device AI Models	26%
■ Communication Module	18%
■ Hardware Extension (Optional)	12%
■ Others & Contingency	10%

FEASIBILITY(Implementation)**Platform Feasibility**Runs on standard Android phones
Using Kotlin/Java and Android Audio APIs.**Model Feasibility**Uses open-source AI4Bharat models
(IndicConformerASR, IndicF5) with sherpa-
Onn and ONN Runtime.**Communication Feasibility**Works with Bluetooth SPP and Wi-Fi Direct for
local, offline, low-bandwidth text transmission.**Offline Feasibility**All processing on-device with local storage.
No cloud APIs required.**Extension Feasibility (Planned)**LoRa/ ESP32 can be integrated for longer-
range communication (1-5 km/
10-20 km) in future versions.**VIABILITY(Real-World Impact)****Social Impact**Enables communication for rural communities,
disaster-affected populations, and non-English
speakers. Supports 10 Indian Languages.**Applicable Environments**Works in remote, disaster-prone areas where
cellular networks and internet are unavailable.**Practical Use Cases**Everyday communication, emergency/SOS alerts,
field coordination, rescue operations, and local
community networks.**Cost-effective and Scalable**Uses open-source models and standard Android
devices. Can be easily distributed at low cost
without dependence on cloud infrastructure.**Sustainable and Long-term**Open-source, offline-first and reusable architecture
ensures long-term usability and minimal
operational cost.**CHALLENGES AND SOLUTIONS****Noisy Environments****Challenge:** Background noise in disaster
or outdoor areas may affect
VAD and ASR accuracy.**Solution:** Use Silero VAD and noise-robust
ASR models with audio preprocessing and
filtering.**Device Fragmentation****Challenge:** Different Android devices
and audio hardware may affect
performance.**Solution:** Test across wide range of
devices, use ONN Runtime and model
quantization for compatibility.**Connection Limitations****Challenge:** Bluetooth/Wi-Fi Direct has
limited range and may face connection
drops or interference.**Solution:** Implement retry mechanisms,
message acknowledgements and optional
LoRa/ESP32 extension.**User Familiarity and Adoption****Challenge:** Some users may be unfamiliar
with push-to-talk interface and voice-
based communication.**Solution:** Simple and intuitive UI with clear
visual feedback, SOS button and minimal
steps for easy adoption.

IMPACT AND BENEFITS

IMPACT



Disaster & low-connectivity communication

Works when cellular towers or internet infrastructure are unavailable. Enables **offline, device-to-device** communication.



Multilingual inclusion

Voice-first communication across **10 Indian languages** reduces language and literacy barriers.



Bandwidth resilience

Speech becomes **compact text** before transmission, reducing dependence on continuous audio and enabling low-bandwidth communication.



Emergency communication

Push-to-talk relay and SOS alerts provide rapid communication **PTT + SOS** channels fail.



BENEFIT



Social

More **inclusive** voice-based communication for people facing language or literacy barriers.



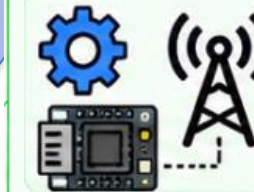
Connectivity

Phone-to-phone text relay uses **Bluetooth SPP / Wi-Fi Direct** without cellular towers or cloud APIs.



Offline resilience

On-device ASR/TTS and local communication continue **without internet or cloud services**.



Strategic

Open-source, lightweight architecture reduces dependence on proprietary cloud services and supports future **LoRa/ESP32 extension**.

KEY RESEARCH PAPERS & REFERENCES



Cyclone Fani 2019 (NIDM)

A case study on Cyclone Fani, preparedness, evacuation and communication challenges

[Click Here](#)


Hanspal & Behera (IJDRM 2024)

Role of Emerging Technology in Disaster Management in India

[Click Here](#)


Disaster Management in India - A Status Report (2004)

Ministry of Home Affairs, Govt. of India (Hazard exposure and communication)

[Click Here](#)


Disaster Management in India (UNDP, 2004)

Same MHA status report (hosted by UNDP)

[Click Here](#)


Charkhi Dadri Mid-air Collision (1996)

Example of language barrier in safety-critical communication

[Click Here](#)


Manipur Earthquake 2016 (NIDM)

A case study on Manipur Earthquake (M6.7, 4 Jan 2016)

[Click Here](#)


Kotrupi Landslide 2017 (NIDM)

A case study on Kotrupi landslide, Mandi district, Himachal Pradesh

[Click Here](#)


Bihar Floods 2007 (NIDM)

A case study on Bihar floods 2007 (impact and communication systems)

[Click Here](#)

KEY DATA & INSIGHTS FROM RESEARCH



~40 M ha

Flood-prone land in India, about 1/8 of the country (NIDM, Bihar Floods 2007)



~60%

of India's landmass is earthquake-prone, 68% drought-prone, 8% cyclone-prone (MHA Status Report 2004)



20 M people, 510 deaths

Affected by the Bihar floods 2007. Power and phone lines were cut in most flood-hit areas, but mobile phones kept working



1,2 M evacuated in 24 h

Cyclone Fani 2019. 64 people died, against over 10,000 in the 1999 super cyclone (NIDM / Prevention Web)



5-layer comms failed

All five layers of Odisha's satellite and terrestrial communication system failed soon after Fani made landfall (NIDM / Prevention Web)



84%

Of Mandi district's population is rural (site of the Kotrupi landslide, 46 dead) (NIDM, Kotrupi 2017)

"Communication is the first casualty in a calamity, so multi-mode, redundant communication was decided on."

- MHA, Disaster Management in India: A Status Report (2004)

COMPARISON WITH EXISTING SOLUTIONS

Features	GENERIC ASR/TTS Apps	Emergency Radio Systems	VaaniSetu
Works offline	×	⚠️	✓
Low-bandwidth	×	⚠️	✓
Text transmission	×	⚠️	✓
10 Indian languages	×	×	✓
Push-To-Talk (PTT)	⚠️	✓	✓
Distress/SOS alerts	×	✓	✓
No cloud APIs	×	✓	✓
Optional long-range (LoRa/ESP32)	×	⚠️	✓

EVIDENCE FROM REAL INCIDENTS

Events	Communication Gap	VaaniSetu Angle
Bihar floods 2007	Landslides and power down, Phones still worked	Phone-to-phone works when infrastructure fails
Cyclone Fani 2019	All 5 communication layers failed after landfall	Needs no tower or cloud
Kotrupi landslide 2017	Warning relied on home guards and a megaphone. IoT early-warning system was vandalized more than once.	Low-cost, device-based alerts
Charkhi Dadri 1996	A mid-air collision traced to weak English on ATC radio.	Multilingual voice for safety-critical talk



OFFICIAL GUIDELINES & GOVERNMENT REFERENCES

MHA, Disaster Management in India: A Status Report (2004)

[Click Here](#)


NIDM Case Studies (Bihar 2007, Kotrupi 2017, Cyclone Fani 2019, Manipur 2016)

[Click Here](#)


Hanspal & Behera (IJDRM 2024) Role of Emerging Technology in Disaster Management in India

[Click Here](#)


PROJECT RESOURCES

GitHub Repository

[Click Here](#)


Architecture Document

[Click Here](#)


Demo Video

[Click Here](#)